

Review of Proposed changes of selectivity patterns in the North Sea sole SS3 assessment ICES WGNSSK 2026

Hans Gerritsen – Marine Institute – Ireland – version 1 – 22/04/2026

Background

The current Stock Synthesis assessment for North Sea sole uses a random walk approach to fitting selectivity, which is also time-varying for the landings fleet. Landings were estimated to have full selectivity for older ages up to 2022 but in the last 2 years the model estimated a dome-shaped selection pattern. This could lead to the creation of cryptic biomass in the stock in the future if the landings selectivity does not revert to a logistic curve. Moreover, the selection pattern for the surveys appears to be unrealistic.

Because of these issues, WGNSSK have investigated the impact of reducing the flexibility by substituting the random walk for logistic selectivity for the landings and double-normal for the other fleets. Additionally, the time-varying landings selectivity has been restricted to the years 2009 to 2021 only, which represents the period between the introduction and the complete ban of pulse trawling.

Review

It is my opinion that:

- The proposed change is well justified
- The perception of the stock is not meaningfully altered (SSB and recruitment differences are almost imperceptible and differences in the time series of F are minor, considering the uncertainty bounds)
- Residuals of commercial catch numbers-at-age are larger for young ages but this is expected behaviour, due to reduced flexibility. I found no significant changes to other diagnostics
- Because the changes have not meaningfully changed the estimated stock development, reference points are likely unaffected. Nevertheless, if it is straightforward to re-estimate reference points to check that these are essentially unchanged this would add further confirmation that the current reference points remain valid.

